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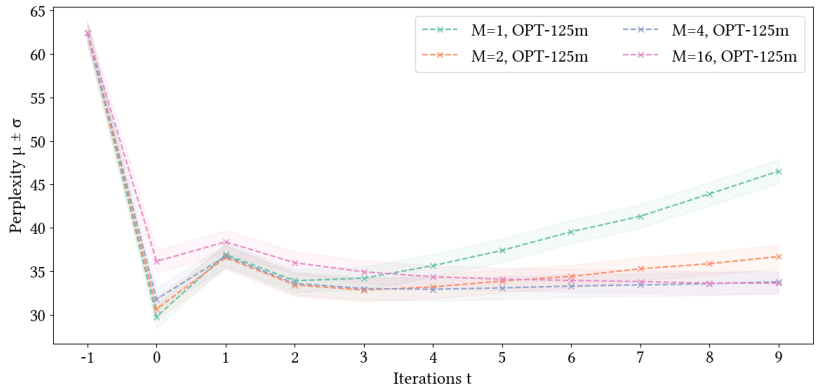
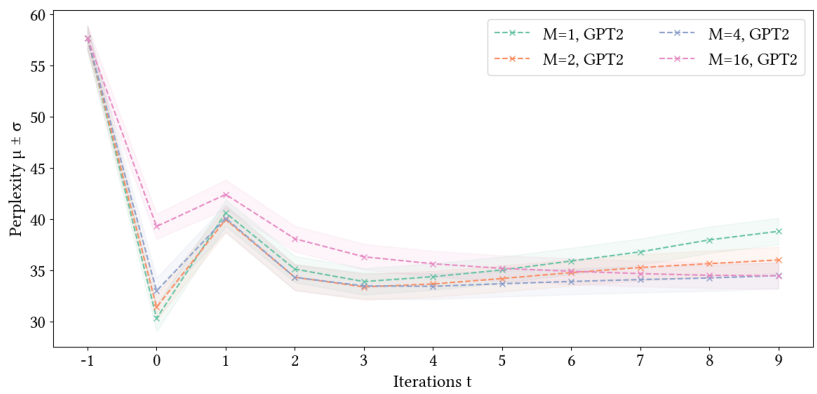
A Model progress including performance on base models

Figure 8 shows the perplexity progress of the two model families across ten iteration steps, including the models' perplexity for the base models.



(a) OPT-125m



(b) GPT2

Fig. 8. **Comparison of ecosystems' perplexity.** In this figure, the comparison is made by increasing diversity ($D = M$) across ten iteration steps, including models' perplexity for the off-the-shelf models ($t = -1$). The significant perplexity drop at the first iteration step shows that the fine-tuning mechanism works.

B OPT-125m M=1 runs with different random seeds

Figure 9 shows the performance trajectories for $M = 1$ across ten iterations for four independent runs. The similarity of these trajectories indicates that the results are largely independent of the random seed. Ecosystems with higher diversity already average performance across two or more models.

C OPT-125m performance across 20 iterations

Figure 10 shows the optimal diversity level for OPT-125m across up to 20 self-training iterations (T), showing that the optimal diversity level continues to increase beyond 10 iterations.

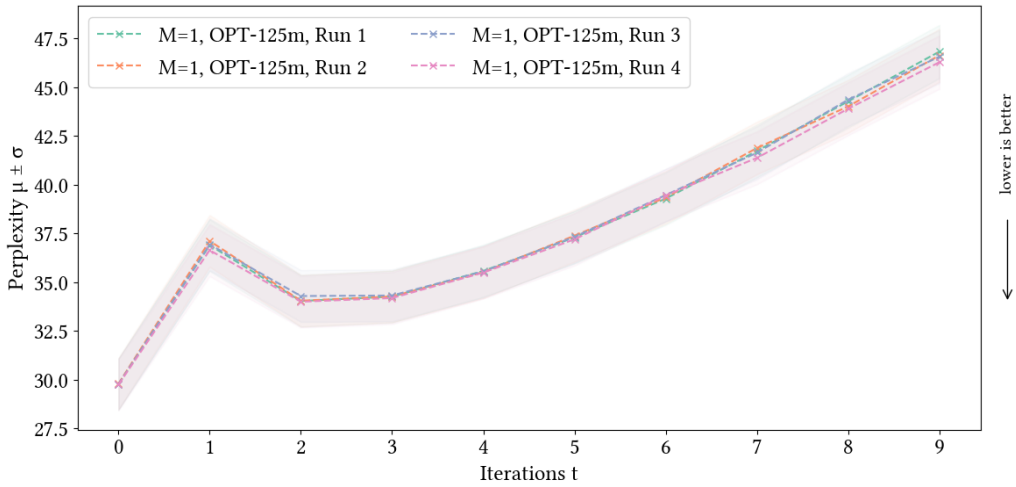


Fig. 9. **The performance trajectory of $M = 1$ is largely run-independent.** Mean perplexity and standard deviation for ecosystem $M = 1$ for each of four independent runs with different random seeds.

D Illustrations of probability distributions and EDQ

Figure 11 illustrates the probability distributions in self-training loops, and Figure 12 illustrates the model-dependent EDQ.

E Perplexity distributions for both model families

Figure 13 shows the mixed perplexity distributions of the OPT-125m and GPT2 ecosystems on the original training dataset, indicating that model collections with greater diversity exhibit both greater variety in the information represented and lower alignment with the training data distribution.

F GPT2 V1: Model size and data size

Figure 14 shows GPT2 performance results for increased model size (GPT2-Medium) and larger data size (2.1M tokens).

G Additional experimental variations of the primary setting

The observed short-term benefit of homogeneous ecosystems is due to the segmentation approach, which reduces the data size per model as diversity increases, and to the use of a universal ground-truth dataset (the Wikitext2 test set in our case). This initial advantage disappears when diversity is instantiated through combining two model families (Figure 16), through data expansion (Figure 17b), or when community-specific test sets are employed. The aggregated results are shown in Figure 15.

Diversity through combining 2 model families. Real-world diversity may arise from multiple system-design dimensions, including pre-training data, architecture, alignment, interface design, and system messaging. Figure 16 shows that combining two model families (OPT-125m and GPT2) consistently outperforms the combination of two models of same family, suggesting that diversity across design dimensions may further amplify benefits.